

ETF proxy definition validation

Factor	Candidate	Corr.	Sign match	Beta	t-stat	R2	Raw TE
SMB	IJR - SPY*	92.3\%	87.2\%	0.828	129.3***	85.3\%	4.6\%
SMB	IJR - QUAL	89.7\%	85.0\%	0.775	109.2***	80.5\%	5.5\%
SMB	IJR - SPHQ	88.9\%	83.5\%	0.740	104.3***	79.0\%	6.0\%
SMB	IJR - QQQ	74.7\%	75.9\%	0.485	60.5***	55.9\%	11.2\%
HML	SPYV - SPYG*	75.5\%	76.3\%	0.904	61.9***	57.1\%	9.2\%
HML	SPYV - QQQ	77.1\%	76.4\%	0.773	65.1***	59.5\%	9.4\%
HML	SPYV - SPY	74.2\%	74.5\%	1.607	59.5***	55.1\%	10.1\%
HML	SPYV - ARKK	55.3\%	68.6\%	0.244	35.7***	30.6\%	26.6\%
RMW	QUAL - ARKK*	65.1\%	69.4\%	0.191	46.1***	42.4\%	24.3\%
RMW	SPHQ - SPHB	29.0\%	64.5\%	0.151	16.3***	8.4\%	16.0\%
RMW	QUAL - SPHB	28.6\%	62.9\%	0.161	16.1***	8.2\%	15.1\%
RMW	QUAL - SPY	16.4\%	56.6\%	0.427	8.9***	2.7\%	8.6\%
CMA	SPHQ - QQQ*	57.3\%	70.2\%	0.438	37.6***	32.8\%	8.4\%
CMA	SPYV - SPYG	53.6\%	69.2\%	0.352	34.1***	28.8\%	10.0\%
CMA	SPLV - QQQ	52.9\%	68.0\%	0.241	33.5***	27.9\%	14.3\%
CMA	SPLV - SPHB	19.8\%	54.3\%	0.068	10.8***	3.9\%	22.1\%

Notes: Asterisk denotes the proxy used in the main empirical analysis.

Validation of ETF proxies against Fama and French factors

Factor	Corr.	Sign match	Beta	t-stat	R2	Raw TE	Residual vol.
SMB	92.3\%	87.2\%	0.828	129.3	85.3\%	4.6\%	4.1\%
HML	75.5\%	76.3\%	0.904	61.9	57.1\%	9.2\%	9.2\%
RMW	65.1\%	69.4\%	0.191	46.1	42.4\%	24.3\%	6.5\%
CMA	57.3\%	70.2\%	0.438	37.6	32.8\%	8.4\%	6.3\%

Primary daily Losers strategy and SPY benchmark

Factor	Strategy	A / B	Acc.	Final	CAGR	Vol.	Sharpe	MDD	Switches	Alpha	Beta
CMA	CMA-13L	SPHQ / QQQ	50.6\%	454.9	14.2\%	20.1\%	0.760	-32.9\%	311	-0.1\%	1.080
HML	HML-13L	SPYV / SPYG	49.4\%	318.0	10.7\%	18.8\%	0.632	-36.9\%	321	-2.4\%	1.008
RMW	RMW-13L	QUAL / ARKK	48.0\%	180.9	5.3\%	31.4\%	0.323	-71.5\%	278	-9.1\%	1.349
SMB	SMB-13L	IJR / SPY	50.2\%	356.5	11.8\%	20.6\%	0.645	-42.5\%	330	-1.5\%	1.040
SPY	Buy and hold	SPY		425.2	13.5\%	17.7\%	0.807	-33.7\%			

Daily Winners and Losers comparison at k=13

Factor	Rule	Accuracy	Final	CAGR	Sharpe	MDD	Switches
CMA	Losers	50.6\%	454.9	14.2\%	0.760	-32.9\%	311
CMA	Winners	49.4\%	581.4	16.7\%	0.897	-30.7\%	311
HML	Losers	49.4\%	318.0	10.7\%	0.632	-36.9\%	321
HML	Winners	50.6\%	458.3	14.3\%	0.826	-31.3\%	321
RMW	Losers	48.0\%	180.9	5.3\%	0.323	-71.5\%	278
RMW	Winners	52.0\%	812.9	20.2\%	0.789	-46.6\%	278
SMB	Losers	50.2\%	356.5	11.8\%	0.645	-42.5\%	330
SMB	Winners	49.8\%	308.7	10.4\%	0.603	-33.7\%	330

Selected classification strategy results

Frequency	Factor	Strategy	Final	CAGR	Vol.	Sharpe	MDD	Switches
Daily	HML	HML-Logit	583.3	18.5\%	19.4\%	0.974	-30.3\%	469
Daily	RMW	RMW-KNN11	617.2	19.2\%	29.1\%	0.748	-43.2\%	920
Daily	CMA	CMA-KNN3	446.6	15.5\%	20.2\%	0.815	-32.8\%	1042
Daily	SMB	SMB-KNN3	313.2	11.6\%	20.2\%	0.646	-35.2\%	1151
Weekly	HML	HML-KNN15	511.2	17.0\%	17.6\%	0.983	-30.9\%	213
Weekly	RMW	RMW-KNN11	701.7	20.6\%	31.9\%	0.747	-68.4\%	233
Weekly	SMB	SMB-KNN23	468.7	16.0\%	19.3\%	0.868	-32.5\%	198
Weekly	CMA	CMA-KNN23	502.4	16.8\%	18.9\%	0.918	-29.9\%	193

Selected strategy performance at 0, 1, and 2 bps per switch

Frequency	Strategy	SPY final	0 bps	1 bps	2 bps
Daily	HML-Logit	410.2	640.7 [†] / 1.02	611.3 [†] / 1.00	583.3 [†] / 0.97
Daily	RMW-KNN11	410.2	741.7 [†] / 0.81	676.6 [†] / 0.78	617.2 [†] / 0.75
Daily	CMA-KNN3	410.2	550.0 [†] / 0.91	495.6 [†] / 0.87	446.6 [†] / 0.82
Daily	SMB-KNN3	410.2	394.2 / 0.76	351.4 / 0.70	313.2 / 0.65
Weekly	HML-KNN15	414.5	533.4 [†] / 1.01	522.2 [†] / 0.99	511.2 [†] / 0.98
Weekly	RMW-KNN11	414.5	735.1 [†] / 0.76	718.2 [†] / 0.75	701.7 [†] / 0.75
Weekly	SMB-KNN23	414.5	487.6 [†] / 0.89	478.1 [†] / 0.88	468.7 [†] / 0.87
Weekly	CMA-KNN23	414.5	522.1 [†] / 0.94	512.1 [†] / 0.93	502.4 [†] / 0.92

Notes: Each strategy cell reports terminal wealth followed by annualized Sharpe ratio. Dagger indicates terminal wealth exceeds SPY over the same sample.

Out of sample strategy performance

Factor	Obs.	Final	CAGR	Vol.	Sharpe	MDD	Switches	Acc.
CMA	752	200.6	26.3\%	17.1\%	1.453	-20.3\%	111	50.0\%
HML	251	97.0	-3.0\%	17.3\%	-0.087	-19.3\%	32	51.0\%
RMW	251	91.6	-8.4\%	23.4\%	-0.258	-28.9\%	31	47.8\%
SMB	582	174.4	27.2\%	19.4\%	1.340	-22.5\%	80	52.6\%

Single proxy regression estimates

Factor	Term	Coef.	Std. Err.	t-Stat	R2	Obs.
SMB	Intercept	0.00001	0.00005	0.17	85.3\%	2889
SMB	SMB proxy	0.82846	0.00641	129.29	85.3\%	
HML	Intercept	0.00015	0.00011	1.43	57.1\%	2889
HML	HML proxy	0.90358	0.01459	61.93	57.1\%	
RMW	Intercept	0.00014	0.00008	1.91	42.4\%	2889
RMW	RMW proxy	0.19120	0.00415	46.07	42.4\%	
CMA	Intercept	0.00005	0.00007	0.66	32.8\%	2889
CMA	CMA proxy	0.43838	0.01167	37.58	32.8\%	

Notes: The dependent variable is the daily Fama French factor return. Conventional OLS standard errors.

Transaction cost sensitivity: final wealth and Sharpe ratio

Strategy	0 bps	2 bps	5 bps	10 bps
Daily CMA Losers	484.1 / 0.787	454.9 / 0.760	414.4 / 0.719	354.7 / 0.651
Daily HML Logit	640.7 / 1.021	583.3 / 0.974	506.8 / 0.904	400.8 / 0.787
Weekly HML KNN	533.4 / 1.006	511.2 / 0.983	479.6 / 0.948	431.3 / 0.889
Weekly SMB KNN	487.6 / 0.887	468.7 / 0.868	441.7 / 0.838	400.2 / 0.788

Weekly KNN strategy performance, best k per factor

Factor	Strategy	A/B	Acc.	Final	CAGR	Vol.	Sharpe	MDD	Switches	Alpha	Beta
CMA	CMA-KNN23	SPHQ / QQQ	48.3\%	502.4	16.8\%	18.9\%	0.918	-29.9\%	193	1.7\%	1.038
HML	HML-KNN15	SPYV / SPYG	52.0\%	511.2	17.0\%	17.6\%	0.983	-30.9\%	213	2.5\%	0.978
RMW	RMW-KNN11	QUAL / ARKK	53.0\%	701.7	20.6\%	31.9\%	0.747	-68.4\%	233	2.6\%	1.405
SMB	SMB-KNN23	IJR / SPY	52.4\%	468.7	16.0\%	19.3\%	0.868	-32.5\%	198	1.0\%	1.042

Notes: Best k is selected by full sample Sharpe ratio at 2 bps per switch.

Daily and weekly logistic regression strategy performance

Freq.	Factor	A/B	Acc.	Final	CAGR	Vol.	Sharpe	MDD	Switches	Alpha / Beta
Daily	CMA	SPHQ / QQQ	49.8\%	460.3	15.8\%	20.9\%	0.808	-36.0\%	533	0.2\% / 1.098
Daily	HML	SPYV / SPYG	53.1\%	583.3	18.5\%	19.4\%	0.974	-30.3\%	469	3.2\% / 1.028
Daily	RMW	QUAL / ARKK	50.6\%	427.1	15.0\%	31.9\%	0.599	-69.0\%	656	-1.0\% / 1.322
Daily	SMB	IJR / SPY	49.4\%	227.8	8.3\%	19.8\%	0.500	-41.7\%	570	-5.7\% / 1.029
Weekly	CMA	SPHQ / QQQ	48.3\%	471.3	16.1\%	18.9\%	0.887	-32.4\%	172	1.1\% / 1.035
Weekly	HML	SPYV / SPYG	51.9\%	335.4	12.4\%	17.3\%	0.762	-33.6\%	165	-1.3\% / 0.958
Weekly	RMW	QUAL / ARKK	48.0\%	269.4	10.0\%	32.3\%	0.456	-60.5\%	169	-6.4\% / 1.397
Weekly	SMB	IJR / SPY	51.3\%	368.5	13.4\%	20.7\%	0.710	-38.3\%	171	-2.1\% / 1.109

Sensitivity of KNN to the number of neighbors k: Sharpe ratio

Freq.	Factor	3	5	7	9	11	13	15	17	19	21	23	25
Daily	CMA	0.82	0.77	0.78	0.69	0.70	0.65	0.61	0.66	0.63	0.69	0.71	0.60
Daily	HML	0.64	0.65	0.80	0.78	0.76	0.81	0.86	0.75	0.82	0.85	0.78	0.76
Daily	RMW	0.55	0.59	0.65	0.66	0.75	0.72	0.62	0.58	0.63	0.60	0.71	0.57
Daily	SMB	0.65	0.53	0.56	0.63	0.55	0.50	0.56	0.51	0.53	0.57	0.49	0.58
Weekly	CMA	0.81	0.83	0.84	0.80	0.82	0.91	0.90	0.90	0.89	0.86	0.92	0.92
Weekly	HML	0.85	0.73	0.74	0.82	0.95	0.95	0.98	0.93	0.78	0.76	0.81	0.85
Weekly	RMW	0.63	0.71	0.53	0.66	0.75	0.65	0.63	0.58	0.63	0.63	0.57	0.50
Weekly	SMB	0.57	0.64	0.78	0.81	0.76	0.72	0.81	0.84	0.77	0.78	0.87	0.82

Notes: Values are annualized Sharpe ratios net of 2 bps per switch.

Payoff asymmetry and directional accuracy in selected strategies

Strategy	Accuracy	mu+	mu-	p*	Timing	PT Stat. (p)
Daily CMA Losers (k = 13)	50.6\%	0.450\%	0.476\%	51.4\%	-0.39	1.06 (0.144)
	Daily HML Logit	53.1\%	0.546\%	0.536\%	1.90	2.05 (0.020)
	Daily HML KNN15	51.9\%	0.546\%	0.536\%	1.25	1.05 (0.146)
	Daily RMW KNN11	51.7\%	1.400\%	1.403\%	2.34	1.72 (0.042)
	Daily CMA KNN3	50.5\%	0.471\%	0.468\%	0.33	0.26 (0.397)
	Daily SMB KNN3	49.9\%	0.586\%	0.562\%	0.57	-0.24 (0.593)
Weekly HML Winners (k = 1)	52.3\%	1.150\%	1.106\%	49.0\%	3.70	0.81 (0.208)
Weekly CMA Losers (k = 1)	50.4\%	1.008\%	1.018\%	50.2\%	0.22	0.51 (0.304)
	Weekly CMA KNN23	48.3\%	1.060\%	0.970\%	0.61	-1.36 (0.913)
	Weekly HML KNN15	52.0\%	1.231\%	1.111\%	5.38	0.68 (0.249)
	Weekly RMW KNN11	53.0\%	3.114\%	2.991\%	12.25	1.17 (0.122)
	Weekly SMB KNN23	52.4\%	1.297\%	1.163\%	6.32	1.06 (0.144)

Notes: mu+ and mu- are average absolute ETF proxy spreads in correctly and incorrectly classified periods.